

BHUMIKA SAXENA

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TECHNICAL SKILLS

Languages: Java, Python, C/C++ (fundamentals), TypeScript, JavaScript, SQL

Core CS: Data Structures & Algorithms, Object-Oriented Design, Operating Systems, Computer Networks, System Design, Complexity Analysis

Backend & Databases: FastAPI, Node.js, Express.js, Next.js, REST APIs, Celery, Redis, PostgreSQL, MongoDB, Cloudinary

Machine Learning: PyTorch, XGBoost, Scikit-learn, Sentence Transformers, Pandas, NumPy

Tools: Git, Docker, GitHub Actions, Postman, Vercel, Railway, VS Code

EXPERIENCE

Software Developer Intern — Black Cube Solutions, Dubai (Remote) Jan 2026 – Jun 2026

Node.js • Express.js • Next.js • TypeScript • PostgreSQL • MongoDB • REST APIs • Git

- Shipped 50+ production REST APIs across 3 enterprise systems as a core engineer in a 2–3 person team, owning design through deployment.
- Designed auth and RBAC from ambiguous requirements, enabling role-based workflows across all platforms.
- Designed PostgreSQL schemas from scratch for 3 products and optimized MongoDB queries, improving data integrity and performance under high-frequency load.
- Built the Class Plus frontend end-to-end in Next.js/TypeScript — from database to UI.
- Coordinated across 11 repositories with continuous code review and debugging over a 6-month engagement.

PROJECTS

Token Engine — LLM Context Optimization Framework 2026

Python • FastAPI • Groq API • Sentence Transformers • Docker • MCP • GitHub Actions

- Built a framework-agnostic context optimization engine for LLM apps — semantic chunking, embedding-based retrieval, and token-budget management.
- Cut LLM token usage by 65.5% at 470ms end-to-end latency using semantic ranking and an adaptive task planner.
- Architected a decoupled interface layer (FastAPI, MCP, LangChain, CrewAI) atop a single core engine — added Claude Desktop/Cursor MCP support without touching the optimizer, embeddings, or providers.

Async Transaction Processing Pipeline — AI-Powered Financial Data System 2026

Python • FastAPI • Celery • Redis • PostgreSQL • Docker • Gemini API

- Built an async transaction pipeline with FastAPI, Celery, and Redis, returning non-blocking job responses while workers processed financial datasets through cleaning, anomaly detection, and LLM classification.
- Implemented persistent job-state tracking in PostgreSQL, batched Gemini classification, and exponential-backoff retries to handle API rate limits and failures.
- Analyzed high-load scaling bottlenecks across storage and worker memory, designing mitigations with AWS S3 for object storage, PgBouncer for connection pooling, and Redis caching.

Apex Intelligence — AI-Powered Formula 1 Race Intelligence Platform 2026

Python • TypeScript • Next.js • FastAPI • XGBoost • Docker

- Built a full-stack F1 race intelligence platform with an XGBoost pipeline trained on 2020–2025 race, driver, and circuit data.
- Engineered a counterfactual What-If Simulator re-simulating race outcomes with grid changes, DNFs, and weather, computing a Chaos Index per scenario.
- Built a five-pillar driver Skill Score system and an explainability layer exposing feature importance and per-circuit model confidence.

EDUCATION

Bachelor of Technology in Computer Science and Engineering 2023 – 2027

IMS Engineering College, Ghaziabad | CGPA: 8.5 / 10

LEADERSHIP & ACTIVITIES

- President, GeeksforGeeks Student Chapter — Organized coding contests, technical workshops, and peer-learning sessions on data structures, algorithms, and software development.
- Smart India Hackathon 2025 Grand Finalist — Developed a disaster response platform and represented the institute at the national grand finale.